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RV COLLEGE OF ENGINEERING®
 (An Autonomous Institution Affiliated to VTU)
 IV Semester B. E. Examinations Sept./Oct. – 2024
 Civil Engineering

ENVIRONMENT AND SUSTAINABILITY (BASKET COURSE)

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

M BT CO

1	1.1	List the components of environment.	02	1	1
	1.2	Differentiate between primary and secondary succession.	02	4	1
	1.3	Define climate change.	02	1	2
	1.4	List any two renewable and non renewable sources of energy.	02	1	2
	1.5	Define GDP as an economic indicator.	02	1	3
	1.6	Enlist any four Millennium Development Goal (MDG's).	02	1	3
	1.7	List any four ozone depleting substances.	02	1	2
	1.8	Define carbon foot print for a product or process.	02	1	3
	1.9	Write the principles of CSR.	02	2	4
	1.10	What are the two examples of business ethics?	02	2	4

PART-B

2	a	Classify different types of ecosystems. Explain in brief any two ecosystems with their structure and composition.	08	2	1
	b	Briefly describe the various Environmental protection acts promulgated in India.	08	2	2
3	a	Demonstrate on "Hydrogen energy is a clean energy", write the advantages and disadvantages of Hydrogen energy as an energy source.	08	2	2
	b	Enumerate the process of energy management and energy conservation with its objectives.	08	2	2
OR					
4	a	Explain : i. Biomass energy ii. Ocean thermal energy conversion	08	2	2
	b	Define Energy cycle and explain with a neat sketch carbon cycle.	08	2	2
5	a	Discuss Environmental Economics and Environmental audit as a measure to sustainability.	08	2	3
	b	Discuss linear resource management systems as bane in sustainable Growth of industries.	08	2	3
OR					

6	a	Illustrate how sustainable development is concept of three aspects of society.	08	2	2
	b	Define industrial ecology and explain the strategies to implement the same.	08	2	2
7	a	Enumerate all the 17 SDG's stipulated by United nations and brief the significance of any one.	08	1	3
	b	Explain Zero waste concept as a philosophical approach.	08	2	3
OR					
8	a	Describe Circular Economy and distinguish linear vs circular economy.	08	2	3
	b	Explain ISO 14000 and carbon Sequestration.	08	2	3
9	a	Recall and explain the history of evolution of CSR.	08	2	4
	b	Illustrate the concept of charity and corporate philanthropy.	08	2	4
OR					
10	a	Discuss the concept of sustainability and stakeholder Management.	08	2	4
	b	Enumerate the key features of Global Reporting Index and Dow Jones sustainability index.	08	1	4

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RV COLLEGE OF ENGINEERING®
 (An Autonomous Institution Affiliated to VTU)
 III Semester B. E. Examinations April/May-2024

Common to All

ENVIRONMENT AND SUSTAINABILITY

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

M BT CO

1	1.1	Define the terms Biotic and Abiotic components.	02	1	1
	1.2	Mention threats to bio-diversity.	02	1	1
	1.3	List out the types of land pollution.	02	1	2
	1.4	State the effects of noise pollution.	02	1	2
	1.5	What are fossil fuels and why they are non-renewable?	02	1	2
	1.6	What is biogas and mention its uses.	02	1	2
	1.7	Highlight the significance of GDP in nation's economic activity.	02	2	3
	1.8	Highlight the distinction between Kyoto protocol and Paris agreement.	02	2	3
	1.9	Mention the principles of organic farming.	02	1	3
	1.10	Enumerate the significance of EIA in developmental projects.	02	1	4

PART-B

2	a	Discuss the importance of any four environmental protection acts promulgated in India.	08	2	1
	b	Briefly discuss the causes, effects and preventive measures for soil and air pollution.	08	2	2
3	a	Explain the concept of energy management and conservation.	08	2	2
	b	Enumerate the concept of sustainable urbanization and the need for the same.	08	1,2	3
		OR			
4	a	Explain the concept of hydrogen energy with its uses, advantages and disadvantages.	08	2	2
	b	Explain the carbon cycle with sketch.	08	2	2
5	a	Define environmental audit. Briefly describe the types of environmental audit.	08	1	3
	b	Enumerate the economic and social aspects of sustainability.	08	1	3
		OR			

6	a	Explain the key concepts of system thinking along with its advantages.	08	2	3
	b	Discuss the concepts of circular economy and industrial ecology.	08	2	3
7	a	Enumerate the various ways of reducing carbon footprint in our daily life.	08	1	3
	b	State and explain how green building concept evolved as sustainable habitat.	08	2	3
OR					
8	a	Enumerate ISO 14000 series of standards.	08	1	3
	b	Explain in brief the necessity of energy efficiency and sustainable transport.	08	2	3
9	a	Explain the principles of CSR.	08	2	4
	b	Explain the concept of charity and corporate philanthropy.	08	2	4
OR					
10	a	Enumerate the environmental aspects of CSR.	08	1	4
	b	Explain how the changing emphasis of companies focusing on CSR.	08	2	4

**Department of Civil Engineering**

Date	20/6/2024	Maximum Marks	50
Course Code	CV242TA	Duration (minutes)	90
Sem	IV Semester	CIE-I	
ENVIRONMENT AND SUSTAINABILITY			

NOTE: All questions are compulsory.

TEST - I				
QNo.	Questions	M	BT	CO
1.a	Differentiate between natural and artificial ecosystems, providing examples for each	5	2	1
1.b	Describe the energy flow in an ecosystem, emphasizing the role of photosynthesis and nutrient cycles	5	3	1
2.a	Discuss the impact of human activities on biodiversity loss	6	3	3
2.b	What are the main objectives of environmental Protection acts?	4	2	3
3.a	Compare and contrast primary and secondary succession, highlighting their similarities and differences in terms of initiating factors, rate of ecosystem development, and species diversity	5	3	1
3.b	Distinguish between primary and secondary pollutants of air with examples	5	2	2
4.a	Discuss the importance of biodiversity and explain types of biodiversity	10	3	1
5.a	Describe the measures to be taken to control and prevent the following types of Pollution i. Air Pollution ii. Water Pollution	10	3	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	25	15	10			15	35	-	-	-



DEPARTMENT OF CIVIL ENGINEERING

Date	8 th Jan 2024	Maximum Marks	50
Course Code	CV232AT	Duration (minutes)	90
Sem	III Semester	CIE – I	
Environment and Sustainability			

No.	Questions	M	BT	CO
1	a) Define the terms i) Environment ii) Ecology iii) Bio-Diversity	6	1	1
	b) Materials flow in eco system is cyclic and Energy flow is unidirectional, justify.	4	2	1
2.	Classify different types of ecosystems. Explain in brief any two ecosystems with their structure and composition	10	2	2
3.	a) Recall the concept of ecological succession and classify the same	5	1	1
	b) Discuss the approaches of conserving the biodiversity	5	2	1
4.	a) Highlight the salient features of various hotspots of biodiversity in India	5	1	1
	b) Briefly discuss the causes, effects and preventive measures of water pollution	5	2	2
5.	Briefly describe the various Environmental protection acts in India	10	2	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	25	25	--	-	16	34	-	-	-	-



Department of Civil Engineering

Date	23/7/2024	Maximum Marks	50
Course Code	CV242TA	Duration (minutes)	90
Sem	IV Semester	CIE-II	
ENVIRONMENT AND SUSTAINABILITY			

NOTE: All questions are compulsory.

QNo.	Questions	M	BT	CO
1.	The last two decades shows an alarming increase in global temperatures. The rise has been directly linked to the increase in _____	2	1	2
2.	Write two disadvantages of Tidal energy	2	1	2
3.	What is meant by Carbon Sequestration	2	1	3
4.	Write two objectives of fair trade	2	1	1
5.	How Corporate Social Responsibility leads to sustainability.	2	2	3

TEST - I				
QNo.	Questions	M	BT	CO
1.	Brief on i) Bio mass energy ii) Ocean Thermal Energy conversion	10	1	1
2.	Demonstrate on "Hydrogen energy is a clean energy" write the advantages and Disadvantages of Hydrogen energy as a energy source.	10	3	3
3.	Write on four approaches and principles of Green engineering.	10	1	3
4.	Explain the importance of Environmental auditing towards the sustainable growth	10	2	2
5 a	Explain Source and sink in Carbon cycle.	05	2	1
5. b	Explain the prescriptive based and incentive based Environmental Economics.	05	2	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	15	10	25		20	20	10	-	-	-



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Department of Civil Engineering

Academic year 2023-2024 (Odd Sem)

Date	20-02-2024	Maximum Marks	50
Course Code	CV232AT	Duration	90 Min
Sem	III (Basket Course)	CIE – II (Test)	
ENVIRONMENT AND SUSTAINABILITY			

Sl. No.	Questions	Marks	BT	CO
1.	Enumerate the process of energy management and energy conservation with its objectives.	10	2	3
2.	Explain any 5 types of alternate energy sources with their advantages and disadvantages.	10	3	1
3.	Enumerate the objectives of sustainable urban projects.	10	2	3
4.	Highlight five important principles of sustainable development	10	3	3
5.	a. Briefly explain any 5 international protocols and agreements that address specific environmental issues.	05	2	2
	b. Describe any 5 disadvantages of linear resource management systems.	05	3	2

Course outcomes:

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	10	30	-	-	25	25	-	-	-



Department of Civil Engineering

Date	27/08/2024	Maximum Marks	10Q + 50T
Course Code	CV242TA	Duration (minutes)	120
Sem	IV Semester	IMPROVEMENT CIE	
ENVIRONMENT AND SUSTAINABILITY			

NOTE: All questions are compulsory.

IMPROVEMENT QUIZ				
Q. No.	Questions	M	BT	CO
1.1	Mention any two major causes for climate changes.	2	1	3
1.2	List the benefits of circular economy.	2	1	3
1.3	Discuss energy efficiency, and why is it important in reducing environmental impact?	2	2	3
1.4	List the advantages of Green building.	2	1	4
1.5	Discuss the challenges, when trying to implement the 5Rs?	2	2	4

IMPROVEMENT TEST				
Q. No.	Questions	M	BT	CO
1.	Explain the key objectives of conducting an Environmental Impact Assessment (EIA).	10	2	3
2.	Discuss the role of Life Cycle Assessment (LCA) in calculating the carbon footprint of a product.	10	2	3
3.	Explain the principles of Corporate Social Responsibility.	10	2	4
4.	Discuss the history and evolution of Corporate Social Responsibility (CSR).	12	2	4
5.	Discuss the benefits of corporate philanthropy.	08	2	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	-	26	34	06	54	-	-	-	-
